

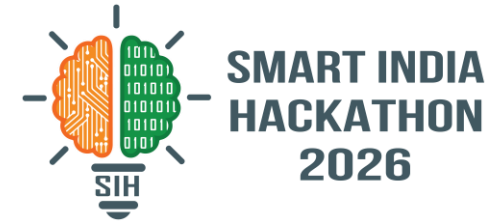
FieldTest Recorder

- **Problem Statement ID** – SIH26231
- **Problem Statement Title** – Digital Companion for Field Drug Testing
- **Theme** – MedTech / BioTech / HealthTech
- **PS Category** – Software
- **Team ID** – 152041
- **Team Name** – DARTH CodersUNI



FieldTest Recorder

A guided, measured and signed field test for NCB's Narcotic Drugs Detection Kit



THE PROBLEM IT ADDRESSES

- **Read by eye.** Officers match the kit's colour against a printed chart; nothing proves a test happened at a place and time (NCB, this problem statement).
- **No standard.** Bombay HC (2021): NCB "has not prescribed the standards"; field testing is "arbitrary", and a bare mention in the panchnama is not enough.
- **Wrong drug, wrong quantity.** Kerala HC (2024, Anuraj): seized "MDMA" was methamphetamine, which changes the NDPS quantity category.
- **So:** disputed seizures, bail on procedure, and field results that cannot later be checked against the lab report.

PROPOSED SOLUTION (PHONE APP, OFFLINE)

- 1 Guide**
NCB kit Tests A–E in the order of the kit's printed flow charts, with the drops and steps printed for each test
- 2 Capture**
Photo of the reaction on a reference colour card beside a reagent-only blank well (spot-plate tests first; tube holder for Tests B and E next)
- 3 Read**
Colour bands from the kit's printed ranges: positive / no colour change / inconclusive, plus every drug on the kit chart that the colour fits
- 4 Sign**
Time, GPS, officer and image hash in every record, signed and hash-chained (designed for the phone's secure hardware key)
- 5 Anchor**
A 14-character code written into the witness-signed panchnama
- 6 Verify**
Anyone can check the records offline with our verifier. Planned: Field Test Memo, Form-1 item 5 text, BSA s.63 data

INNOVATION AND UNIQUENESS

Says only what the chemistry supports

Test E blue = "cocaine or methaqualone" until E3/E4 narrows it; warns when the NDPS quantity category can't be told in the field.

Designed for varying sample amounts

SIMULATED

In simulation, colour bands + a blank well gave 0% false positives where our first single-point rule gave up to 11.7%. Real reactions: next.

Proves completeness, not just integrity

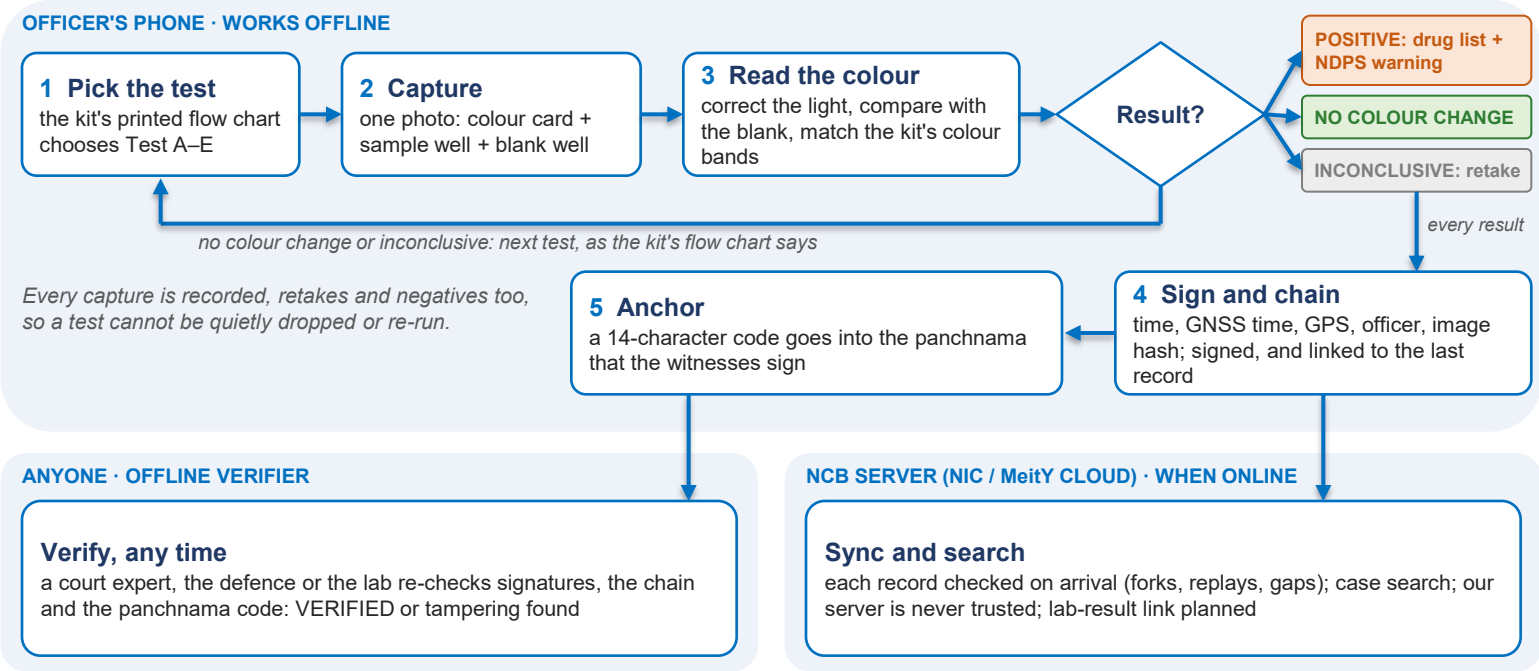
PROVEN IN CODE

Once the panchnama code is written, deleted or re-run tests are caught, and a server admin cannot forge a record (prototype, test keys).

No new hardware

Existing NCB kit + officer's phone + a printed card. Designed to work offline and to be embeddable in apps such as e-Sakshya.

PROCESS FLOW: ONE PACKAGE, FROM SEIZURE TO COURT



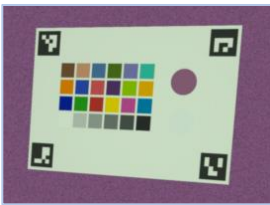
Every step above runs today in our Python prototype (colour on simulated photos). The Android app and hardware-backed keys are next.

Trust: NCB signs the supervisor list; a supervisor's key (a Class 3 DSC in production) binds officer + phone + key; the phone key is designed to live in StrongBox/TEE (prototype: test keys).

Technologies to be used: Kotlin · CameraX/Camera2 RAW · OpenCV · Android Keystore + attestation · SQLCipher · RFC 8785 JSON · ECDSA P-256 · Spring Boot · PostgreSQL · S3. Prototype today: Python, OpenCV, SQLite.

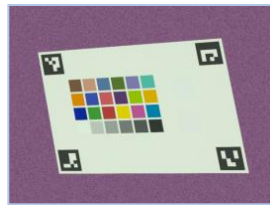
Deliberately not used: blockchain, cloud AI, a neural-network "drug detector" (a colour test cannot identify a drug; the court must be able to re-run the reading).

WORKING PROTOTYPE OUTPUT



POSITIVE

Test A: fits codeine, heroin, morphine



NEGATIVE

Test A: no colour change vs blank



INCONCLUSIVE

Test C: kit chart not printed blank

Real reference code on camera-model photos: finds the card, corrects the light, reads the well against the blank, decides, then signs and chains the record; the raid verifies offline.

IMPLEMENTATION STATUS

PROVEN IN CODE

Python reference prototype + one-command demo, 65 automated test cases: github.com/Harshith029/fieldtest-recorder

SIMULATED

Colour, liveness, night-light and amount experiments on simulated photos

NEXT

Android app; printed-card and real-reaction validation with NCB/CFSL chemists

18/18

PROVEN IN CODE

of our tampering attacks caught by the verifier, incl. insider-admin forgery and deleted records

15/15

PROVEN IN CODE

sync and failure scenarios handled by the reference sync protocol, incl. an old-backup restore

11.7% → 0%

SIMULATED

false positives as sample amount varies: our first single-point rule vs colour bands (published US colours)

≤ 1%

NEXT

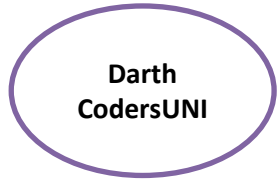
false-positive target for real-kit validation: 1,296 captures, 3 sample amounts, 95% confidence

POTENTIAL CHALLENGES, RISKS AND STRATEGIES

Challenge / risk	Strategy
Real reaction colours differ from the printed chart	Validation by NCB/CFSL chemists under their licences: 1,296 captures at 3 sample amounts; we never handle narcotics
Phones and light vary; glare; night raids	Reference card + RAW capture and a bad-light gate (prototype); say "inconclusive" rather than guess; flash/no-flash night mode (simulated, not yet integrated)
Staged or swapped sample; contaminated blank	A blank that shows a reaction colour gives inconclusive, never negative (prototype). Planned: capture from before the drop, liveness check (simulated so far), package label in frame, s.105 video link
Rooted phone, insider, deleted records	Supervisor-signed binding, panchnama anchor, untrusted server (prototype); hardware keys + attestation (designed; checker prototyped)
Access to SIMS / e-Sakshya not confirmed	File exports first; the engine is designed as an SDK that e-Sakshya could embed

FEASIBILITY ANALYSIS

- **Existing kit, existing phones:** NCB already supplies the kit free; the additions are a printed colour card per kit and a few reagent drops for the blank well.
- **Small infrastructure (estimate):** one app server pair, PostgreSQL, 1–3 TB a year of storage on NIC or MeitY-empenned cloud.
- **Government-owned by design:** NCB would hold the signing keys, the colour standard and the code; open formats, no per-device licence fees.
- **Pass criteria set in advance:** false positives at most 1%, proven at 95% confidence (at least 299 negative controls); "inconclusive" at most 15%.
- **Clear status:** every number is labelled proven in code, simulated or next. No real phone photo or real reaction has been tested yet; that is the next stage.



IMPACT AND BENEFITS



POTENTIAL IMPACT ON THE TARGET AUDIENCE

- SO

Seizing officer
A guided test and a signed record that shows good-faith action; planned: Field Test Memo and Form-1 item 5 filled in.
- IO

Investigating officer, prosecutor
Checkable records for remand and bail; planned: a Rule 10(2) grouping check and the lab result linked back.
- CT

Courts, defence, the accused
Any expert can re-check the record offline; the quantity category is flagged as uncertain until the lab confirms.
- HQ

NCB headquarters and labs
Planned: field-vs-lab accuracy per kit batch and per test, from lab reports the IO already receives.

WHAT CHANGES FOR ONE SEIZURE

Today (paper)	With FieldTest Recorder
Colour judged by eye	Measured against a signed colour standard
"Dark brown" written in the panchnama	Photo + reading + every drug on the kit chart the colour fits
No proof of time or place	Signed time, GNSS time, GPS and officer
Records can be rewritten later	Deletions and re-runs are detectable
Quantity category guessed	Flagged when it can't be told in the field
No kit accuracy data	Field result vs lab result, per kit batch (planned)

Why a field report must not overstate: a real case

Anuraj v. State of Kerala (2024): police alleged commercial-quantity MDMA; the lab found methamphetamine in intermediate quantity (MDMA 0.5 g / 10 g vs methamphetamine 2 g / 50 g). Our record states only what the kit's colour supports, lists every drug on the kit chart that the colour fits, and warns when they fall under different NDPS thresholds.

Social

Fairer remand and bail decisions; fewer cases lost on procedure; honest officers protected.

Economic

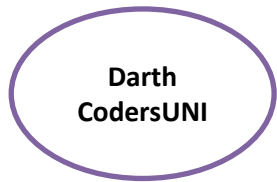
No new hardware; reuses the free NCB kit and officers' phones; small hosting cost.

Environmental

Paperless records; the only extra consumables are a printed card per kit and blank-well reagent drops.

Scale

1,15,236 NDPS cases in 2022 (MHA); each seized package can get a recorded field test.



RESEARCH AND REFERENCES



[1] **SIH26231 problem statement** — Narcotics Control Bureau, MHA; sih.gov.in

[2] **NCB Narcotic Drugs Detection Kit** — NICFS, A Forensic Guide for Crime Investigators, ch. 8, Figs 8.12–8.16 (Tests A–E, flow charts)

[3] **NDPS (Seizure, Storage, Sampling and Disposal) Rules 2022** — GSR 899(E): Rules 3, 5, 10(2), 13, 14

[4] **Sagar Parshuram Joshi v. State of Maharashtra** — Bombay HC, 15 Jan 2021 (indiankanoon.org/doc/90970274)

[5] **Masibur Khan v. State** — Delhi HC, 31 May 2023 (indiankanoon.org/doc/91398841)

[6] **Anuraj v. State of Kerala** — Kerala HC, 2024: alleged MDMA found to be methamphetamine

[7] **NDPS small and commercial quantities** — S.O. 1055(E), 19 Oct 2001

[8] **NCB Drug Law Enforcement Field Officers' Handbook** — narcoticsindia.nic.in

[9] **BPR&D SOP: audio-video recording under BNSS; BSA 2023 s.63** — hash value in the electronic-record certificate

[10] **NIJ Standard-0604.01** — Color Test Reagents/Kits for Preliminary Identification of Drugs of Abuse (US DoJ)

[11] **Android key attestation; RFC 8785 (JSON canonicalisation)** — [developer.android.com; rfc-editor.org](https://developer.android.com/rfc-editor.org)

[12] **MHA Rajya Sabha USQ 1832 (11 Dec 2024)** — NDPS case statistics

OUR CODE AND RESEARCH: github.com/Harshith029/fieldtest-recorder

- **Working Python reference prototype with a one-command demo and 65 automated test cases:** record format and offline verifier, colour engine, NCB kit protocol, searchable log, sync protocol. Photos in the demo are simulated; the Android app must match its outputs.
- **17 experiments with 95% confidence intervals**, including 18 tampering attacks and 15 failure scenarios (real code) and colour, sample-amount, night-light and staged-sample tests (simulated photos).
- **Reviewed adversarially, claim by claim:** confirmed defects are either fixed and re-tested or listed openly in the repository's problem register.
- **Safety and law:** our team never handles narcotics; colour tests use published colours, the kit chart and safe dyes. Real-kit validation is designed for NCB or CFSL chemists.